9/22/2026

HTML

Full-Stack Development Lab

Instructor: Sir M. Fahad Qasim

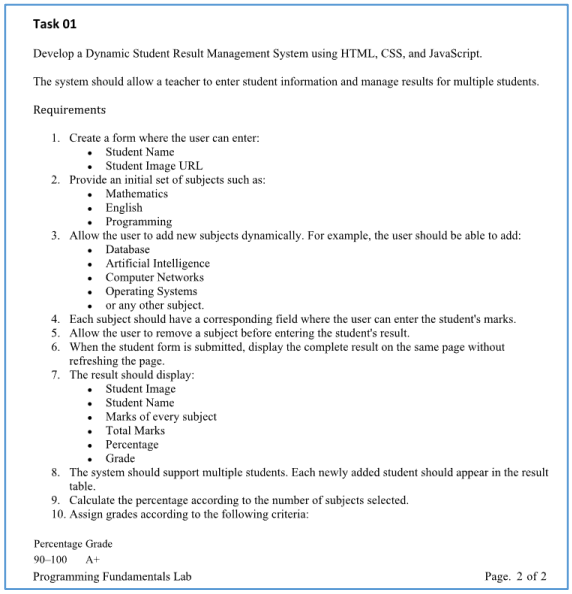
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Class: BSCS F24 B



Lab No 02



Task 01

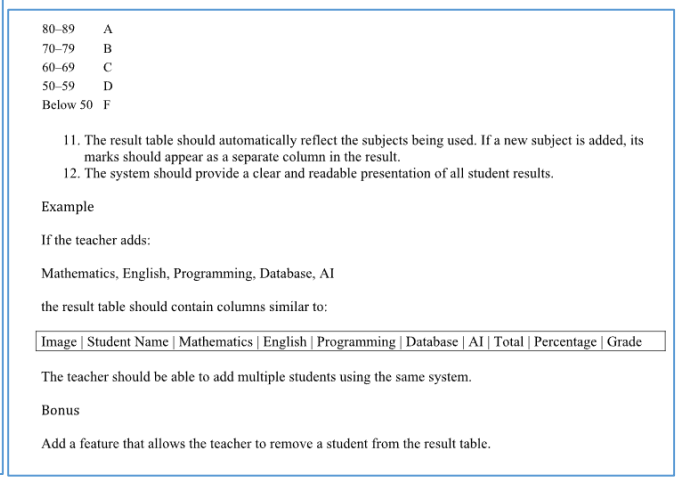
Develop a Dynamic Student Result Management System using HTML, CSS, and JavaScript.

The system should allow a teacher to enter student information and manage results for multiple students.

Requirements

1. Create a form where the user can enter:
   - Student Name
   - Student Image URL
2. Provide an initial set of subjects such as:
   - Mathematics
   - English
   - Programming
3. Allow the user to add new subjects dynamically. For example, the user should be able to add:
   - Database
   - Artificial Intelligence
   - Computer Networks
   - Operating Systems
   - or any other subject.
4. Each subject should have a corresponding field where the user can enter the student's marks.
5. Allow the user to remove a subject before entering the student's result.
6. When the student form is submitted, display the complete result on the same page without refreshing the page.
7. The result should display:
   - Student Image
   - Student Name
   - Marks of every subject
   - Total Marks
   - Percentage
   - Grade
8. The system should support multiple students. Each newly added student should appear in the result table.
9. Calculate the percentage according to the number of subjects selected.
10. Assign grades according to the following criteria:

Percentage Grade
90–100 A+
Programming Fundamentals Lab Page. 2 of 2



80–89 A
70–79 B
60–69 C
50–59 D
Below 50 F

11. The result table should automatically reflect the subjects being used. If a new subject is added, its marks should appear as a separate column in the result.
12. The system should provide a clear and readable presentation of all student results.

Example

If the teacher adds:

Mathematics, English, Programming, Database, AI

the result table should contain columns similar to:

Image	Student Name	Mathematics	English	Programming	Database	AI	Total	Percentage	Grade

The teacher should be able to add multiple students using the same system.

Bonus

Add a feature that allows the teacher to remove a student from the result table.

Code:

```html
<!DOCTYPE html>
<html>

<head>
    <title>Student Result System</title>
    <style>
        body {
            font-family: Arial, sans-serif;
            margin: 20px;
        }

        h2 {
            background-color: #f0f0f0;
            padding: 10px;
            font-weight: normal;
            border-radius: 4px;
        }

        input,
        button {
            padding: 6px;
            margin-right: 10px;
            margin-bottom: 15px;
```

```
            border: 1px solid #a9a9a9;
            border-radius: 2px;
        }

        .sub-row {
            margin-bottom: 10px;
        }

        .sub-row span {
            display: inline-block;
            width: 120px;
        }

        .sub-row input {
            width: 80px;
            margin-bottom: 0;
        }

        .sub-row button {
            padding: 4px 10px;
        }

        table {
            width: 100%;
            border-collapse: collapse;
            margin-top: 10px;
        }

        th,
        td {
            border: 1px solid #a9a9a9;
            padding: 8px;
            text-align: center;
        }

        th {
            background-color: #f9f9f9;
            font-weight: normal;
        }
    </style>
</head>

<body>

    <h2>Student Result System</h2>
```

```
<form id="myForm">
    <div>
        <input id="stdName" required placeholder="Student Name">
        <input id="stdPic" type="url" required placeholder="Image URL">
    </div>

    <div>
        <input id="newSub" placeholder="New Subject Name">
        <button type="button" id="addBtn">Add Subject</button>
    </div>

    <p><b>Enter Marks:</b></p>
    <div id="subInputs"></div>

    <br>
    <button type="submit">Submit Your Result</button>
</form>

<table>
    <thead id="headRow"></thead>
    <tbody id="bodyRows"></tbody>
</table>

<script>
    let mySubjects = ["Math", "English", "Programming"];
    let myStudents = [];

    function updateScreen() {
        let formDiv = document.getElementById("subInputs");
        let tHead = document.getElementById("headRow");
        let tBody = document.getElementById("bodyRows");

        formDiv.innerHTML = "";
        for (let i = 0; i < mySubjects.length; i++) {
            formDiv.innerHTML += `
        <div class="sub-row">
            <span>${mySubjects[i]}:</span>
            <input type="number" id="mark_${i}" required>
            <button type="button" onclick="deleteSub(${i})">x</button>
        </div>`;
        }

        tHead.innerHTML = `<tr><th>Image</th><th>Name</th>`;
        for (let i = 0; i < mySubjects.length; i++) {
```

```
                tHead.innerHTML += `<th>${mySubjects[i]}</th>`;
            }
            tHead.innerHTML +=
`<th>Total</th><th>%</th><th>Grade</th><th>Action</th></tr>`;

            tBody.innerHTML = "";
            for (let i = 0; i < myStudents.length; i++) {
                let std = myStudents[i];
                let rowHtml = `<tr>
                <td><img src="${std.pic}" width="30"></td>
                <td>${std.name}</td>`;

                for (let j = 0; j < mySubjects.length; j++) {
                    let m = std.marks[mySubjects[j]];
                    if (m == undefined) m = "-";
                    rowHtml += `<td>${m}</td>`;
                }

                rowHtml += `<td>${std.totalMarks}</td>
                <td>${std.percentage}</td>
                <td>${std.grade}</td>
                <td><button type="button"
onclick="deleteStd(${i})">x</button></td>
            </tr>`;

                tBody.innerHTML += rowHtml;
            }
        }

        document.getElementById("addBtn").addEventListener("click", function () {
            let val = document.getElementById("newSub").value;
            if (val != "" && mySubjects.includes(val) == false) {
                mySubjects.push(val);
                document.getElementById("newSub").value = "";
                updateScreen();
            }
        });

        document.getElementById("myForm").addEventListener("submit", function
(event) {
            event.preventDefault();

            let studentMarks = {};
            let total = 0;
```

```
        for (let i = 0; i < mySubjects.length; i++) {
            let m = Number(document.getElementById("mark_" + i).value);
            studentMarks[mySubjects[i]] = m;
            total = total + m;
        }

        let per = total / mySubjects.length;

        let g = "F";
        if (per >= 90) { g = "A+"; }
        else if (per >= 80) { g = "A"; }
        else if (per >= 70) { g = "B"; }
        else if (per >= 60) { g = "C"; }
        else if (per >= 50) { g = "D"; }

        let newStudent = {
            name: document.getElementById("stdName").value,
            pic: document.getElementById("stdPic").value,
            marks: studentMarks,
            totalMarks: total,
            percentage: per.toFixed(1),
            grade: g
        };

        myStudents.push(newStudent);
        document.getElementById("myForm").reset();
        updateScreen();
    });

    function deleteSub(index) {
        let tempArray = [];
        for (let i = 0; i < mySubjects.length; i++) {
            if (i != index) tempArray.push(mySubjects[i]);
        }
        mySubjects = tempArray;
        updateScreen();
    }

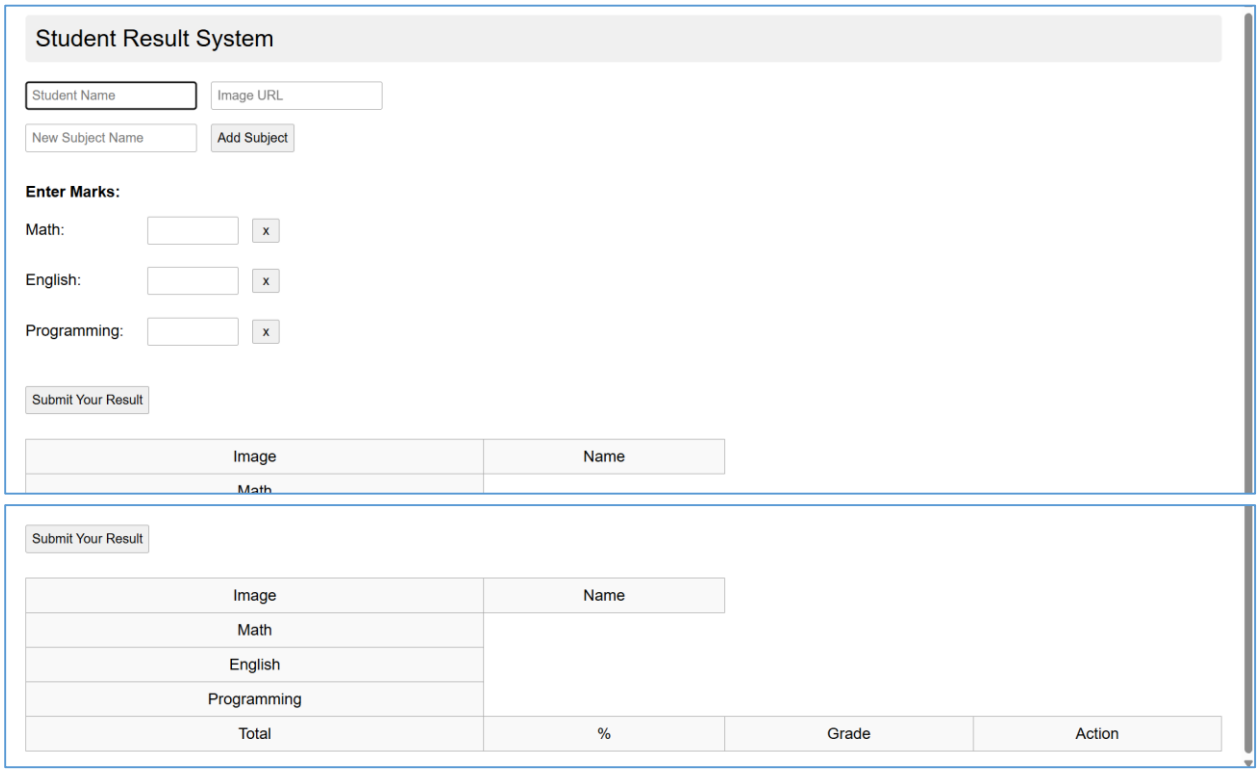
    function deleteStd(index) {
        let tempArray = [];
        for (let i = 0; i < myStudents.length; i++) {
            if (i != index) tempArray.push(myStudents[i]);
        }
        myStudents = tempArray;
        updateScreen();
```

```
        }

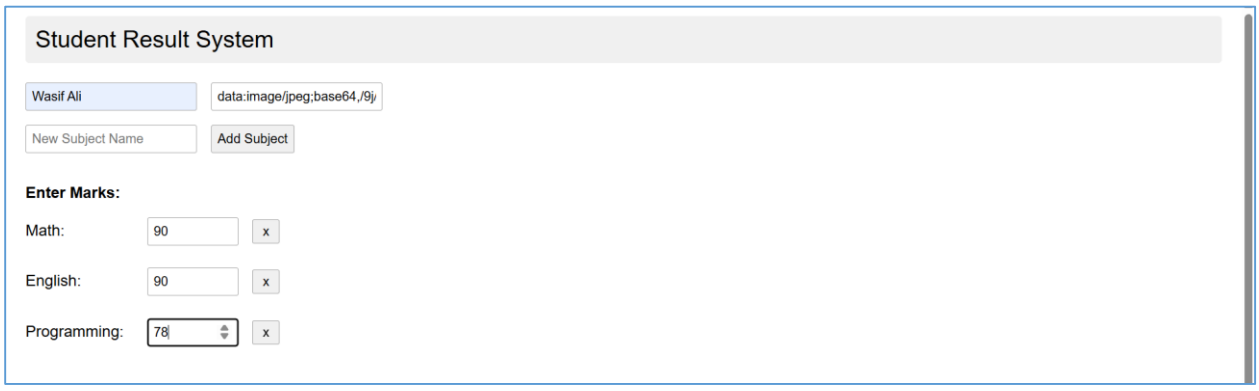
        updateScreen();
    </script>
</body>

</html>
```

OUTPUT:



After adding records:



Submit Your Result

Image	Name
Math	
English	
Programming	

Total	%	Grade	Action					
	Wasif Ali	90	90	78	258	86.0	A	x